Subject Code : 23A01703a

GROUND IMPROVEMENT TECHNIQUES

UNIT - IV

Soil Stabilisation: Mechanical, Cement, Lime, Bituminous and Chemical

Syllabus topics covered in this unit

- Methods of Stabilisation
- Mechanical Stabilisation
- Cement Stabilisation
- Lime Stabilisation
- Bituminous Stabilisation
- Chemical Stabilisation with Calcium Chloride, Sodium Silicate and Gypsum

How to use these notes

- Red words = keywords to remember for the exam.
- Green words = examples, units and values.
- Yellow boxes = quick points; pink boxes = important questions / formulae.
- Learn the sketches - they fetch marks in the answer paper.

1. Soil Stabilisation - Introduction

Soil stabilisation is the process of *improving the engineering properties* of the soil - strength, durability, volume stability, permeability and resistance to the water and the weather - by *mechanical, chemical or physical means*, so that it can be used for the roads, foundations, embankments and the earth structures.

Objectives

- Increase the *shear strength* (CBR, UCS) and the *bearing capacity*; reduce the *compressibility and settlement*.
- Reduce the *plasticity* and *swell-shrink* (volume change) of the expansive clays.
- Reduce the *permeability* and the water absorption; improve the *durability* (freeze-thaw, wet-dry), reduce the *dust* and the erosion.
- Use the *local soil* instead of the imported material - *economy* and sustainability (reduce thickness of the pavement by 30 - 50 %).

Methods of stabilisation

Group	Methods
Mechanical	Compaction, soil blending (mixing of the different gradations), the grading, soil replacement
Chemical / admixture	Cement, lime, lime-fly ash, bitumen (asphalt), calcium chloride, sodium chloride, sodium silicate, gypsum, polymers, enzymes, resins, industrial wastes (fly ash, slag, rice husk ash, GGBS)
Physical	Heating, freezing, drainage and dewatering, electro-osmosis, grouting, stone / lime columns
Reinforcement / geosynthetic	Fibres (natural / synthetic), geotextile, geogrid, geocell, reinforced earth
Biological / new	Biocementation (MICP - Microbially Induced Calcite Precipitation), bio-polymers, enzymes

Soil type	Best stabiliser
Gravel and sand (granular, low fines)	Cement, bitumen, mechanical stabilisation
Clayey sand, silty sand, PI < 20	Cement, lime + fly ash
Clay (PI > 20), expansive soil	Lime, lime + cement, fly ash
Organic soil, peat	Cement + additives (or replace), pre-load
Dust control on gravel / unpaved roads	$CaCl_2$, NaCl, bitumen emulsion, lignosulphonate

Factors affecting the choice

* Soil type and gradation (% fines, PI), the purpose (road base, sub-grade, foundation), the climate (wet / frost), the availability and cost of the stabiliser, the construction equipment, the durability needed.

2. Mechanical Stabilisation

In mechanical stabilisation the properties of the soil are improved by changing its gradation (grain size distribution) and by compaction, without the use of the chemical admixture. It is the oldest and the cheapest method.

Principle

- A well graded soil (gravel - sand - fines) has a dense packing (the voids of the large particles filled by the smaller particles); this gives the high density, strength and low permeability, and the fine fraction supplies the cohesion (binding).
- The required gradation is obtained by blending (mixing) two or more soils (e.g., the coarse sand + the clay / the gravel + the fine soil) in the correct ratio; the plasticity should be moderate: PI 4 - 9 for the base and up to 15 for the sub-base.

Blending - the proportions

Rothfuchs / trial method

* If soils A (coarse) and B (fine) are mixed in the proportion $x : (1 - x)$ the grading of the mix at any sieve = $x\,p_A + (1 - x)\,p_B$ (p = % passing); fit it to the specification envelope (IRC / MoRTH / IS 2720).
* Example: A has 20 % passing 75 micron, B has 90 %; the target is 35 %. Then $20\,x + 90\,(1 - x) = 35 \Rightarrow x = 0.786$ (78.6 % A, 21.4 % B).

- Rothfuchs method (the triangle graph) and the graphical / trial and error methods are used for the mix design; the Proctor test gives the MDD and OMC; the CBR and the Atterberg limits check the quality.
- Typical specification of the stabilised WBM / granular sub-base (GSB): gravel 20 - 50 %, sand 30 - 60 %, fines 8 - 20 %, LL < 25 - 35, PI < 6 - 10 (IRC 37, MoRTH Clause 401 - 405).

Construction steps

Scarify and pulverise the soil → Add the blend soil and water to the OMC → Mix by the grader / pulvi-mixer → Compact by the roller (95 - 100 % MDD)

- Compaction - the static, vibratory and pneumatic tyre rollers; the layer thickness 100 - 200 mm; 95 - 98 % of the modified Proctor density; water content OMC +/- 2 %.
- Other mechanical methods: soil replacement, proof rolling, drainage (side drains, crown to give a 3 - 4 % slope), geosynthetic inclusion.

Advantages	Limitations
Cheap, simple equipment, no chemical, the local material	Strength gain is limited; sensitive to the moisture - it softens in the rain
The soil is reusable, quick	The fines should be controlled; not for the highly plastic clay or the heavy traffic on its own

3. Cement Stabilisation

Soil-cement is the mixture of pulverised soil, Portland cement and water, compacted at OMC and cured, which hardens into a rigid, durable material. Introduced in the USA in 1935; widely used for road bases, sub-bases, the canal lining, the slope protection of the dams and the foundation of the buildings.

Mechanism

- Hydration of the cement: tricalcium silicate (C_3S) and the dicalcium silicate (C_2S) react with the water to form the calcium silicate hydrate (C-S-H) gel and the calcium hydroxide $Ca(OH)_2$ - the cementing bonds between the soil particles.
- In clays the $Ca(OH)_2$ released also gives cation exchange, flocculation and the pozzolanic reaction with the clay silica / alumina - additional strength and the reduction of the plasticity.
- The strength increases with the cement content and the time (curing: 7, 28, 90 days).

Soils and the cement content

Soil (AASHTO)	Cement content by weight of the dry soil (%)
A-1-a (gravel, well graded)	3 - 5
A-1-b, A-3 (sand)	5 - 9
A-2 (silty / clayey gravel and sand)	5 - 9
A-4, A-5 (silt)	8 - 12
A-6, A-7 (clay)	10 - 15
Typical rule	Suitable for the soil with PI < 20 - 30 and fines < 35 - 50 %; the highly plastic clay and organic soil are not suited as the pulverisation is difficult

Mix design - the steps (IS 4332 / PCA method)

Classify the soil; assume the cement content → Find the OMC and the MDD of the soil-cement (IS 4332 Part 3) → Prepare the specimens (UCS), cure 7 days → Select the cement for the UCS 1.7 - 3.5 MPa; wet-dry test

- Criteria: UCS at 7 days = 1.7 - 2.5 MPa for the base, 0.7 - 1.5 MPa for the sub-base and the foundation (IRC:SP:89); freeze-thaw and wet-dry durability tests (soil-cement loss < 14 % for A-1 and A-3, < 10 % for the A-4 to A-7).
- Modified soil-cement (the less cement, 2 - 4 %) for the sub-grade modification; cement-treated base (CTB), cement-bound granular material (CBGM), the roller compacted concrete is for heavy duty.
- Construction: pulverise the soil (80 % passing the 4.75 mm sieve), spread the cement, mix (by the pulvi-mixer, the in-situ mix-in-place or the central plant mix), add water, compact within 2 hours, cure by the membrane / sprinkling / bitumen emulsion for 7 days.

Example

* Soil-cement for the sub-base: 5 % cement; the dry soil per m^3 (MDD = 1.9 t/m^3) = 1.9 t; cement = 0.05 x 1.9 = 0.095 t = 95 kg/m^3.

Advantages	Limitations
High strength and durable, a rigid layer, good load spread; reduces the thickness of the pavement	Shrinkage cracks (reflect to the surface) - use the saw cuts, the cement < 6 %
Reduces the plasticity and the swell; good for the sandy soils	Not suited for the organic soil, the sulphates and the high plastic clays; cost; the time limit for the compaction

4. Lime Stabilisation

Lime stabilisation is the addition of the *lime (quicklime CaO or the hydrated lime $Ca(OH)_2$ or the lime slurry)* to the *fine grained soil (clays)*, which reacts with the soil to reduce the plasticity, the swelling and to increase the strength. It is the best method for the *clay soils (PI > 10 - 15)*, including the *black cotton soil*.

Mechanisms (lime-soil reactions)

Reaction	Process	Effect
1. Cation exchange	Ca^{2+} replaces the Na^+, K^+, H^+ on the clay surface	Thinner double layer, *flocculation* and aggregation - reduced plasticity, easier to work with the clay (immediate, in hours)
2. Flocculation and agglomeration	Lime raises the electrolyte concentration	Soil becomes the silt-like and granular
3. Lime carbonation	$Ca(OH)_2 + CO_2 \rightarrow CaCO_3$	Weak cementation (and loss of the lime)
4. Pozzolanic reaction	At the pH > 12.4 the *clay silica and alumina dissolve* and react with the Ca^{2+}: forms *CSH, CAH and CASH (calcium silicate / aluminate hydrates)*	Long term *strength gain* (months to years); the main strength source

- *Quicklime* reacts with the water (*slakes*), generates heat, dries the wet soils (*drying effect*, 1 % lime reduces w by about 1 %). It is used for the wet fine soil.

Quantity of the lime and the mix design

- *Lime content*: 2 - 8 percent by the dry weight (*lime modification 1 - 3 %*: the plasticity reduction; *lime stabilisation 3 - 8 %*: the strength).
- *Initial Consumption of Lime (ICL)* - the *Eades and Grim pH test*: the minimum lime content for the pH of 12.4 in the soil-lime-water mixture.
- The *UCS* target: 0.7 - 1.0 MPa at the 28 days for the sub-base (IRC: SP: 89 / IS 4332 Part 2); the CBR after the 7 days cure + 4 days soak; the plasticity reduction (PI < 10).
- *Lime-fly ash (LFA)*: lime : fly ash = 1 : 3 to 1 : 4 (the pozzolana) with the soil 10 - 20 % - slow strength gain; *lime-cement-fly ash*; *lime-sand* etc.

Construction (lime treated sub-grade)

Pulverise the soil; spread the lime → First mix; mellowing for 24 - 72 h → Remix; add the water to OMC → Compact; cure for 7 days

- The mellowing (mellowing time) lets the lime react with the clay and the lumps break; compaction by sheep-foot / pad-foot, vibratory rollers; wet curing.
- Applications: the sub-grade, sub-base and base of the roads, canal lining and banks, the foundation pads in BC soils, the lime piles and the lime slurry injection (unit I), the embankment on the soft ground, and the quick working platform.

Advantages	Limitations
Reduces the PI and the swell dramatically; good for the heavy clays; the strength rises with time; cheap; fast drying	Not suitable for the sandy / granular soils (no clay to react) and the organic soil
Improved the workability, the shrinkage reduced	Sulphate-rich soils form ettringite / thaumasite and heave - the treatment needs care; the carbonation; the dust / caustic hazard

5. Bituminous Stabilisation

Bituminous stabilisation is the mixing of the bitumen (asphalt) in the form of the cut-back, emulsion or the hot bitumen (including the foamed bitumen) into the soil. The bitumen acts as a binder and a waterproofing agent, so it is applied in the cohesionless soils (sands) and the granular soils for the flexible pavements.

Types and the action

Type	Action / soil	Binder content
Sand-bitumen (sand bitumen stabilisation)	Cohesionless sand: the bitumen coats the sand grains and binds them - gives the cohesion; for the sands, A-3, A-2 with the fines < 12 %	4 - 10 percent
Soil-bitumen (fine soil, waterproofing)	The bitumen coats the soil aggregates (lumps) and prevents the water access; keeps the moisture constant (the cohesive soil with low PI, the sand-clay)	4 - 7 percent of the cut-back / emulsion
Oiled earth (road surface)	Gravel / earth road with the sprayed bitumen on the top	Surface
Bitumen-lime / bitumen-cement	Bitumen emulsion + the 1 - 2 % cement / lime to break the emulsion in clay, the strength	-
Foamed bitumen stabilisation	The hot bitumen + 2 % cold water gives the foam (the volume 10 - 15 times), mixes with the damp, wet soil (cold recycling of the pavement)	2 - 4 percent

- Binder: cut-back bitumen (the bitumen dissolved in kerosene / the petrol - rapid (RC), medium (MC) and slow curing (SC)) for the soil-bitumen; emulsion (cationic or anionic, SS-1 / MS), tar and foamed bitumen (penetration grade 80 / 100 or VG-10 / VG-30).
- Soil suitable: the passing 75 micron < 25 %, LL < 40, PI < 18 (better for the PI < 10) in the soil-bitumen; sand-bitumen is for the dune sand - the arid deserts (Rajasthan) and the beach sand.
- Mix design: the bitumen content by the Hubbard-Field / Marshall type stability (Marshall stability > 3 kN) or the UCS and water absorption, the optimum fluid content (the water + the binder) for the maximum density; the cure.

Construction

Pulverise the soil to the fine → Add the water (pre-wetting) and the bitumen → Mix (pug-mill / in place), aerate to release the solvent → Compact; seal coat, cure

- The mixing is by the road mixer, the pug mill; the aeration to evaporate the kerosene in the cut-back; the curing time is 2 - 7 days; the surface is sealed to prevent the traffic abrasion.
- Applications: the base and the sub-base in the desert, the earth dams (the impervious facing: the asphaltic lining of the canals and reservoirs), the embankment slopes, the airfields, the shoulders, the dust control.

Advantages	Limit
Waterproof, flexible, resists the erosion and the moisture variation; works in the sand	The c high p

6. Chemical Stabilisation: Calcium Chloride

In the chemical stabilisation the small amounts of the chemical (salts, polymers, acids) change the soil-water electro-chemistry, the pore fluid, or form the cementing gel. Used for the dust control, frost resistance, the moisture control and the local strengthening.

Calcium chloride ($CaCl_2$)

- Properties: a hygroscopic salt (absorbs the moisture from the air and the humidity above 29 - 30 %), the deliquescent; a highly soluble; commercial forms flake, pellet, the brine (30 - 35 %).
- Action: (1) the moisture retention - keeps the surface damp and the soil bonded, preventing the dust and the raveling of the unpaved roads; (2) reduces the surface tension and the vapour pressure of the water, so the evaporation is reduced (the soil dries slowly) and the freezing point of the water is lowered (the frost heave is reduced - an anti-freeze); (3) the flocculation by the cation exchange (Ca^{2+}) reduces the plasticity and increases the dry density; (4) the lubricant that helps the compaction (increases the MDD with the lower OMC).
- Dosage: 0.5 - 2 % (by the dry weight of the soil) as the flakes 77 % - the 1 - 1.5 kg/m^2 for the dust control, or the 0.15 - 0.5 % in the base course.
- Applications: the unpaved roads, WBM / gravel roads, the airfield, the soil-cement curing aid (an accelerator), the cold weather concrete / the frozen soil, dust control in the mines.
- Limitations: leaches out with the rain (needs the reapplication), corrodes the metals, the harmful to the vegetation, the cost.

Remember

- $CaCl_2$: hygroscopic + deliquescent + lowers the freezing point + flocculation => used for the dust control and the frost areas. Sodium chloride (NaCl) acts similarly at the dose 1 - 3 % and is the cheaper in the arid regions.

7. Sodium Silicate and Gypsum Stabilisation

Sodium silicate (water glass, Na_2SiO_3)

- Sodium silicate (Na_2O . n SiO_2) is the viscous alkaline liquid (n = 2 - 4; the "silicate modulus"). With the reagent ($CaCl_2$, the salts, the acids, the cement, formamide) it forms the silica gel - the cementing gel (Si - O - Si network) that binds the soil particles and fills the pores.
- Reaction: $Na_2SiO_3 + CaCl_2 \rightarrow CaSiO_3$ (gel) + 2 NaCl; the $CaSiO_3$ / silica gel hardens, reduces the permeability and the strength of the sand rises to 0.5 - 3 MPa. The gel time (seconds to hours) is controlled by the reagent and the dilution.
- Joosten process (the 2-shot): the injection of the sodium silicate and then the $CaCl_2$ with the double tubes, the strong gel (> 3 MPa) but the lateral spread limited; the 1-shot (Rodio / Siroc): the silicate with the delayed reagent (ethyl acetate / formamide) is injected as the single liquid - the controlled gel time; the low viscosity grout (1.5 - 5 cP), the penetrates the fine sands ($K > 10^{-4}$ cm/s).
- Applications: the underpinning, the tunnel and the excavation support in the sand, the seepage cut-off, the foundation strengthening of the old structures, the sand liquefaction prevention; the soil stabilisation in the pavement by the 2 - 5 % with the lime / cement (the alkali activation).
- Limits: the gel is less durable (the syneresis - the shrinkage, the leaching of the soda in the long term), pH high, not suited for the clays and the silt directly.

Gypsum ($CaSO_4$. 2 H_2O)

- Gypsum supplies the Ca^{2+} and the sulphate to the soil: used for the dispersive / sodic soil, the alkaline soil (the sodium rich) and the soft clay and the peat.
- Action: (1) the cation exchange (Ca^{2+} replaces the Na^+) - the flocculation, the reduction in the swell and the dispersion; (2) the slow dissolution (solubility 2 g/l) gives the long term effect; (3) with the cement / lime / fly ash, the sulphate activation helps to form the ettringite (the calcium sulpho-aluminate) that gives the early strength in the clays and the fly ash / slag blends (the gypsum activated slag).
- Dose: the 2 - 5 % (the soil-gypsum), the 5 - 10 % of the phosphogypsum / FGD gypsum (the industrial waste) in the sub-grade and the embankments, the combination 3 % cement + 3 % gypsum.
- Applications: the reclamation of the sodic (alkali) soil (the agriculture), the stabilisation of the dispersive clay in the earth dam, the soft clays (gypsum + fly ash) and the low volume roads, the gypsum plaster for the mud wall (earthen construction).
- Cautions: excessive gypsum causes the swelling by the ettringite (heave) in the stabilised soils; the sulphate attack to the concrete; the gypsum is soluble and loses the effect in the flowing water.

Chemical	Principal action	Best for
$CaCl_2$	Hygroscopic, flocculation, anti-freeze	Dust control, gravel roads, frost areas
NaCl	Moisture retention, the lowered evaporation	Arid roads (cheap)
Sodium silicate	Gel formation (silica), cementation	Sands, underpinning, cut-off
Gypsum	Ca^{2+} source, flocculation, the ettringite	Sodic / dispersive soil, soft clay
Polymers, enzymes, lignosulphonates	Bonding, the reduced water affinity	Roads, dust, special

8. Comparison and Selection of the Stabilisers

Stabiliser	Soils	Content (%)	Strength / effect	Main use
Mechanical	Any, by blending	-	CBR 20 - 80	Base, sub-base
Cement	Sand, silty and the clayey sand, PI < 20	3 - 12	UCS 1.7 - 5 MPa	Road base, the foundation
Lime	Clay, PI > 10 - 15	2 - 8	UCS 0.7 - 2 MPa, slow	Sub-grade, BC soil, canals
Bitumen	Sand, low PI soil	4 - 10	Waterproof, cohesion	Desert roads, the lining
$CaCl_2$	Gravel, the road	0.5 - 2	Dust and moisture control	Unpaved roads
Sodium silicate	Sand ($k > 10^{-4}$)	10 - 25 % volume of the grout	0.5 - 3 MPa	Underpinning, cut-off
Gypsum	Sodic / dispersive clay	2 - 10	Flocculation	Reclamation, dams

Laboratory tests for the stabilised soils

- Compaction (MDD, OMC), UCS (7, 28 days), CBR (soaked), Atterberg limits, durability (the wet - dry, freeze - thaw), free swell, the permeability, the pH (the lime); the IS 4332 (Parts 1 - 9) series - methods of test for the stabilised soils.
- Field control: the field density (the sand replacement), the plate load, DCP (Dynamic Cone Penetrometer), the Benkelman beam and the FWD; the thickness; the cement / lime content by the EDTA titration.

Environmental points and the sustainability

- The use of the industrial waste (fly ash, GGBS, the rice husk ash, the phosphogypsum, the cement kiln dust, the steel slag) as the partial stabiliser reduces the cost and the CO_2 footprint of the cement and the lime (the cement ~ 0.9 t CO_2/t).
- Possible leaching of the chemicals to the ground water (the high pH, the chlorides, the heavy metals) must be checked (the TCLP test).

Quick revision of Unit IV

- Mechanical: blending + compaction, well graded soil. Cement: the hydration, the 3 - 12 %, UCS 1.7 - 5 MPa, PI < 20.
- Lime: the cation exchange + pozzolanic, 2 - 8 %, the best for the clay / BC soil, ICL by the pH 12.4 test.
- Bitumen: sand-bitumen and soil-bitumen, 4 - 10 %, waterproofing. $CaCl_2$: hygroscopic, the dust control; sodium silicate: the silica gel, the Joosten 2-shot; gypsum: Ca^{2+} source, the sodic soil.

Important Questions (JNTUA pattern)

* Explain the different methods of soil stabilisation and the factors governing the choice of the method.
* Describe the mechanical stabilisation of the soil. How is the proportion of the blending determined?
* Explain the mechanism and construction of the cement stabilised soil. Discuss the cement content for the different soils.
* Explain lime stabilisation. Discuss the lime - soil reactions and the mix design (ICL).
* Describe the bituminous stabilisation of the soil with the types of the binders.
* Explain the chemical stabilisation with $CaCl_2$, sodium silicate and gypsum.
* Compare the cement, lime and bitumen stabilisation. Which one is suited for the expansive soil?